

From Scalar Certificates to Spectral Bounds: Radial Pulsations, Discrete Hardy Forms, and $SU(N)$ Lattice Gauge Theory

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Abstract

A scalar inequality becomes a spectral statement only after its operator, norm, domain, and comparison direction have been identified. We carry out that identification for three families of certificates in a public research repository. For adiabatic radial pulsations, an exact Newtonian work identity and a relativistic positive-function comparison distinguish a homologous trial quotient from the fundamental eigenvalue. The first post-Newtonian correction is a positive structure integral; for an $n = 3$ polytrope its coefficient in the repository normalization is approximately 3.37342294. For a specified finite Dirichlet difference form, we obtain the complete sine spectrum and the sharp lower bound uniform in box size, including the shift caused by constant confinement. Direct variation of the proposed scalar source action reveals an obstruction to identifying that form with the physical fluctuation operator. For a normalized $SU(N)$ lattice Hamiltonian, min-max gives an untruncated single-plaquette gap bound. An application of Yarotsky's weak-interaction theorem then gives a gap at least $3g^2/16$, uniformly in N and open volume, at sufficiently large g^2 for each fixed spatial dimension. We state precisely which steps are analytic, externally supplied, numerically checked, or Lean-certified. No stellar mass cutoff, unique physical coupling, or continuum Yang-Mills gap is inferred.

1 Introduction and scope

An algebraic expression can be positive without being an eigenvalue or a lower bound on one. A positive Rayleigh quotient, for example, need not exclude a negative eigenvalue. Similarly, a finite matrix calculation does not by itself bound the spectrum of an untruncated operator, and a sharp inequality for a chosen quadratic form does not identify that form with a physical second variation. These distinctions motivate the present work.

The starting objects are three repository certificates labelled I13, I14, and I15. Their scalar statements concern, respectively, pressure and relativistic work, a Dirichlet difference energy, and an N -dependent strong-coupling expression. The labels are provenance identifiers, not additional assumptions. The purpose here is to give the operator statements that can be supported, and to locate the physical identifications that remain missing. The accompanying derivations and executable records are collected in the source archive described in Section 6.

The methods are established ones: radial variational principles, completion of squares, finite Dirichlet diagonalization, compact-group representation theory, and stability of gapped product systems. We claim no novelty for those methods or for the existence of strong-coupling lattice

gaps. The specific contribution is an explicit normalization and domain analysis connecting them to the stated certificates, together with counterarguments to stronger identifications.

For radial pulsations, the differential equation and density convention are tied to the versioned source of Saio et al. [1]. For lattice gauge theory, the character description is consistent with the Hamiltonian treatment of Ligterink, Walet, and Bishop [3]; the volume-uniform result uses Yarotsky’s theorem [2] as an external input. The discrete form and the scalar source-action comparison are specified directly below.

Certificate	Supported spectral statement	Additional input still required
I13	Upper trial and lower positive-function comparisons for a specified radial operator	Stellar profiles, equation of state, relevant inner boundary, and controlled relativistic evaluation for a mass cutoff
I14	Complete constant-confinement finite spectrum and sharp bound uniform in box size	Physical action-to-form map and an independent marginality condition for selecting a coupling
I15	Single-plaquette gap for $g^2 \geq 2$; open-volume and canonical GNS gap for $g^2 \geq X_d$	Numerical X_d , any extension to the claimed smaller window, and a separate continuum analysis

Table 1: Scope of the three operator comparisons. The constant X_d is finite but is not numerically evaluated in this work.

We use different symbols where the original certificates reused a parameter: $\chi = 2GM/(Rc^2)$ denotes stellar compactness, L the number of edges in a one-dimensional Dirichlet box, and $x = g^2$ the lattice gauge coupling. All finite-dimensional eigenvalues are relative to the explicitly specified norm. In an infinite-dimensional problem, a statement about the bottom of the spectrum does not imply that the bottom is attained unless this is separately assumed or proved.

2 Radial pulsations: trial values and spectral comparisons

2.1 Equilibrium and domain conventions

Let $P(r)$, $\rho(r)$, $m(r)$, and $\Gamma(r)$ denote pressure, density, enclosed mass, and adiabatic exponent in a spherical equilibrium of radius R . Write a radial displacement as $\xi(r)e^{i\Omega t}$ and put $\eta = \xi/r$. Assume $P, \rho, \Gamma > 0$ in the interior, sufficient local regularity for the equations below, a regular center, and a vacuum surface with the natural pressure boundary condition. Integrations by parts are used only for functions for which the displayed endpoint terms vanish. One may first work on admissible smooth test functions and then take the closure in the radial form norm. Whenever the homologous displacement $\eta = 1$ is used, its membership in that domain is required.

We assume the resulting form has a lower-bounded self-adjoint realization. The notation λ_0 initially means its spectral infimum; statements requiring a lowest eigenfunction additionally assume attainment. These conditions matter at singular endpoints. In particular, a star containing a central black hole has an inner-boundary problem that is not covered by the regular-center hypothesis. Rotation, nonadiabatic driving, accretion, and dissipative boundary terms are also outside this model.

2.2 The exact Newtonian work identity

Set $K = 3\Gamma - 4$. The Newtonian radial equation in fractional displacement is

$$(\Gamma P r^4 \eta')' + r^3 (KP)' \eta + \Omega^2 \rho r^4 \eta = 0. \quad (2.1)$$

Define

$$J[\eta] = \int_0^R \rho r^4 \eta^2 dr, \quad (2.2)$$

$$\mathcal{N}_N[\eta] = \int_0^R (\Gamma P r^4 \eta'^2 - r^3 (KP)' \eta^2) dr. \quad (2.3)$$

Multiplying (2.1) by η shows that an eigenfunction has $\Omega^2 = \mathcal{N}_N[\eta]/J[\eta]$. For a general admissible nonzero η , this ratio is a trial quotient.

Proposition 2.1 (Newtonian square identity). *Suppose the boundary terms in (2.3) and $[r^3 KP \eta^2]_0^R$ vanish. With $\delta = \Gamma - 4/3$,*

$$\mathcal{N}_N[\eta] = \int_0^R P \left\{ \frac{4}{3} r^4 \eta'^2 + \delta r^2 (r\eta' + 3\eta)^2 \right\} dr. \quad (2.4)$$

This identity permits spatially varying Γ .

Proof. Integration of the differentiated pressure factor gives

$$\mathcal{N}_N[\eta] = \int_0^R P (\Gamma r^4 \eta'^2 + 2K r^3 \eta \eta' + 3K r^2 \eta^2) dr.$$

Insert $K = 3\delta$ and $\Gamma = 4/3 + \delta$, and complete the square. No derivative of Γ is discarded in this operation. $\square$

Consequently, $\Gamma \geq 4/3$ pointwise implies nonnegativity of this Newtonian form. If $\Gamma > 4/3$ on a set of positive measure, a zero-energy function must have $\eta' = 0$ and then $\eta = 0$. When the spectral bottom is attained, this rules out a zero lowest eigenvalue and gives $\lambda_0 > 0$. Without attainment, positivity on each nonzero vector alone would not imply a positive uniform lower bound. If $\Gamma = 4/3$ identically, admissible homology is an exact zero mode.

Let

$$W_\beta = \int_0^R K P r^2 dr, \quad W_{\text{gr}} = \int_0^R P r^2 dr, \quad J = \int_0^R \rho r^4 dr. \quad (2.5)$$

The homologous trial has

$$Q_N[1] = \frac{3W_\beta}{J}, \quad \lambda_0 \leq \frac{3W_\beta}{J}. \quad (2.6)$$

The I13 scalar expression has Newtonian part W_β/I . Its equality with this trial quotient therefore requires $I = J/3$. If instead an angular factor is included in a conventional modal inertia, the numerator must carry the corresponding factor as well. The factor three in (2.6) cannot be removed by terminology.

There is also a direct test for when homology is an eigenmode. Acting on 1 with the differential operator in (2.3) gives $-r^3(KP)'$. Thus the quotient can be an eigenvalue of that constant function only if

$$-\frac{(KP)'}{\rho r} \text{ is constant almost everywhere.} \quad (2.7)$$

For constant $\Gamma \neq 4/3$, hydrostatic balance $P' = -Gm\rho/r^2$ turns this into constancy of m/r^3 . A nonuniform-density polytrope therefore generally violates the required condition. This is an exact obstruction to a universal identification of the scalar trial formula with the fundamental eigenvalue.

For constant $K > 0$, the same equation supplies a genuine enclosure:

$$K \operatorname{ess\,inf}_{0 < r < R} \frac{Gm(r)}{r^3} \leq \lambda_0 \leq \frac{3KW_{\text{gr}}}{J}. \quad (2.8)$$

Indeed, the gradient part of (2.3) is nonnegative and the remaining part is the weighted integral of KGm/r^3 . For decreasing density, the mean interior density decreases with r , giving the lower endpoint KGM/R^3 .

2.3 The relativistic operator and a lower comparison

Adopt the metric convention

$$ds^2 = -e^{2a(r)}c^2dt^2 + e^{2b(r)}dr^2 + r^2d\Omega_{\mathbb{S}^2}^2, \quad e^{-2b} = 1 - \frac{2Gm}{rc^2}. \quad (2.9)$$

The density convention is the one in Saio et al. [1], equations (2)–(5), without an additional internal-energy correction. This is a specified operator convention, not an assertion that it supplies a complete equation of state for every stellar application. We require $2Gm/(rc^2) < 1$ throughout the equilibrium. The equilibrium identities used below are

$$P' = -(P + \rho c^2)a', \quad a' + b' = \frac{4\pi G r e^{2b}(P + \rho c^2)}{c^4}. \quad (2.10)$$

Set $z = e^{-a}r^2\xi = e^{-a}r^3\eta$ and $\Omega = c\sigma$. Rearrangement of the radial equation gives

$$-(\Pi z')' + Vz = \sigma^2 W z, \quad (2.11)$$

where

$$\Pi = \frac{e^{3a+b}\Gamma P}{r^2}, \quad W = \frac{e^{a+3b}(P + \rho c^2)}{r^2}, \quad (2.12)$$

$$V = \frac{e^{3a+b}}{r^2} \left(\frac{4P'}{r} + \frac{8\pi G e^{2b} P(P + \rho c^2)}{c^4} - \frac{P'^2}{P + \rho c^2} \right). \quad (2.13)$$

In particular, the dimensional variational quotient is

$$Q_{\text{GR}}[z] = c^2 \frac{\int_0^R (\Pi z'^2 + Vz^2) dr}{\int_0^R W z^2 dr}, \quad \lambda_0 \leq Q_{\text{GR}}[z]. \quad (2.14)$$

Proposition 2.2 (Positive-function comparison). *Let $u > 0$ in the interior, with enough regularity for $-(\Pi u')' + Vu$ to be defined, and suppose $[\Pi u' z^2/u]_0^R = 0$ for the test functions considered. Put*

$$q_u = \frac{c^2 [-(\Pi u')' + Vu]}{Wu}.$$

Then $\lambda_0 \geq \operatorname{ess\,inf} q_u$. If u is also an admissible trial function, then

$$\operatorname{ess\,inf} q_u \leq \lambda_0 \leq Q_{\text{GR}}[u]. \quad (2.15)$$

Proof. Write $z = uv$, expand its derivative, and integrate the cross term. This gives the exact identity

$$\int_0^R (\Pi z'^2 + Vz^2) dr - \int_0^R \frac{-(\Pi u')' + Vu}{u} z^2 dr = \int_0^R \Pi u^2 ((z/u)')^2 dr. \quad (2.16)$$

The right side is nonnegative. Divide by $\int Wz^2$ and take the variational infimum. The upper inequality is (2.14). $\square$

Thus a negative trial value proves a negative spectral bottom, and a positive lower endpoint proves positivity. A positive upper endpoint alone proves neither. If q_u is constant, the two endpoints coincide. Endpoint compatibility in (2.16) is a hypothesis of this conclusion, not a formal consequence of choosing $u > 0$.

2.4 A form suitable for computation and expansion

Substituting $z = e^{-a}r^3\eta$ in (2.14), using (2.10), and integrating the derivative $4(e^{a+b}Pr^3\eta^2)'$ yields $Q_{\text{GR}} = \mathcal{N}_{\text{GR}}/\mathcal{D}_{\text{GR}}$, with

$$\begin{aligned} \mathcal{N}_{\text{GR}}[\eta] = & \int_0^R e^{a+b} \left\{ \Gamma Pr^4 \eta'^2 + 2Pr^3(K - \Gamma r a') \eta \eta' \right. \\ & \left. + [3KPr^2 - (6\Gamma + 2)Pr^3 a' - 2Pr^3 b' + ((\Gamma - 1)P - \rho c^2)r^4 a'^2] \eta^2 \right\} dr, \end{aligned} \quad (2.17)$$

$$\mathcal{D}_{\text{GR}}[\eta] = \int_0^R e^{-a+3b} (\rho + P/c^2) r^4 \eta^2 dr. \quad (2.18)$$

The removed endpoint term vanishes under the regular-center and vacuum surface conditions. The derivative form (2.14) and the expanded form (2.17) provide separate assembly routes for a numerical implementation.

Consider a weak-field sequence about a Newtonian equilibrium with $\Gamma - 4/3 = O(\varepsilon)$ and $Gm/(rc^2) = O(\varepsilon)$. At first order, evaluate the leading integrals on the Newtonian background. For homology, the result is

$$\mathcal{N}_{\text{GR}}[1] = 3W_\beta - \frac{\mathcal{H}}{c^2} + O(\varepsilon^2), \quad (2.19)$$

$$\mathcal{H} = \int_0^R \left(8GPrm + 8\pi GPr^4 \rho + G^2 \rho m^2 \right) dr > 0. \quad (2.20)$$

The order term in (2.19) is measured in the pressure-work scale; it is not a supplied numerical error bound. To see the coefficients, use

$$a' = \frac{Gm}{r^2 c^2} + O(\varepsilon^2/r), \quad b' = \frac{G(4\pi r \rho - m/r^2)}{c^2} + O(\varepsilon^2/r)$$

in (2.17). At $\Gamma = 4/3$, the terms linear in a', b' combine as $-8GPrm/c^2 - 8\pi GPr^4 \rho/c^2$. The factor $-\rho c^2 r^4 a'^2$ supplies the remaining term $-G^2 \rho m^2/c^2$. The other products enter at the next order under the stated scaling. Since the numerator starts at first order and $\mathcal{D}_{\text{GR}}[1] = J + O(\varepsilon)$ in the inertia scale, its denominator correction affects the quotient only at second order.

Define the profile-dependent coefficient

$$C_{\text{gr}} = \frac{\mathcal{H}R}{6GMW_{\text{gr}}}, \quad I = J/3, \quad \chi = \frac{2GM}{Rc^2}. \quad (2.21)$$

Then the first-order homologous quotient takes exactly the scalar form

$$Q_{\text{PN}}[1] = \frac{W_\beta - C_{\text{gr}} \chi W_{\text{gr}}}{I}. \quad (2.22)$$

This identification concerns a trial quotient and a controlled-order expansion. It is not exact linear dependence of a physical eigenvalue on compactness. A varying equilibrium sequence also changes its moments and its structure coefficient.

If an explicit bound $|\mathcal{N}_{\text{GR}}[1] - (3W_\beta - \mathcal{H}/c^2)| \leq E$ is available, then $3W_\beta - \mathcal{H}/c^2 + E < 0$ suffices for full-model instability. Otherwise one must evaluate the exact numerator to make such a conclusion at finite compactness. An asymptotic order symbol alone is insufficient near a sign change.

For a uniform-density Newtonian star with $G = \rho = R = 1$, $m = 4\pi r^3/3$ and $P = 2\pi(1 - r^2)/3$. Direct integration gives $W_{\text{gr}} = 4\pi/45$ and $\mathcal{H} = 304\pi^2/315$, hence

$$\frac{\mathcal{H}R}{18GMW_{\text{gr}}} = \frac{19}{42}, \quad C_{\text{gr}} = \frac{19}{14}. \quad (2.23)$$

For the generated $n = 3$ polytrope, the corresponding numerical values are

$$\frac{\mathcal{H}R}{18GMW_{\text{gr}}} \simeq 1.124474311979, \quad C_{\text{gr}} \simeq 3.373422935937. \quad (2.24)$$

The first value agrees, after rounding, with the 1.1245 benchmark in Saio et al. [1]. The factor-three distinction follows from $K = 3\Gamma - 4$. In this normalization the previously used upper endpoint 3.35 is too low for that polytrope, and no profile-independent coefficient band is justified by these two examples.

3 A complete discrete Dirichlet spectrum

Let $L \geq 2$, let $u = (u_1, \dots, u_{L-1}) \in \mathbb{R}^{L-1}$, and set $u_0 = u_L = 0$. Define

$$\begin{aligned} U[u] &= \sum_{n=0}^{L-1} u_n^2, & D[u] &= \sum_{n=0}^{L-1} (u_{n+1} - u_n)^2, \\ A[u] &= \sum_{n=0}^{L-1} (u_{n+1} - \tfrac{3}{2}u_n)^2, & E[u] &= A[u] - \alpha U[u] + \beta \sum_{n=0}^{L-1} q_n u_n^2. \end{aligned} \quad (3.1)$$

Here α, β are real parameters. The interpretation $\alpha = \kappa^2$, $\beta = \mu^2$ is an additional restriction. We study the matrix H defined by $E[u] = u^T H u$ relative to the norm U . The ordinary coordinate Hessian of E is $2H$; this convention has no effect on signs but does affect quoted eigenvalues.

Theorem 3.1 (Constant-confinement spectrum). *If $q_n = q_0$ and $b_0 = 1/4 - \alpha + \beta q_0$, the eigenvalues of H are simple and equal to*

$$\lambda_j = 6 \sin^2 \left(\frac{j\pi}{2L} \right) + b_0, \quad 1 \leq j < L. \quad (3.2)$$

The corresponding eigenvectors are $v_j(n) = \sin(j\pi n/L)$. In particular,

$$\min_{u \neq 0} \frac{E[u]}{U[u]} = 6 \sin^2 \left(\frac{\pi}{2L} \right) + b_0. \quad (3.3)$$

Proof. Telescoping the endpoint values gives

$$2 \sum_{n=0}^{L-1} (u_{n+1} - u_n) u_n = -D[u].$$

Expanding $u_{n+1} - \frac{3}{2}u_n = (u_{n+1} - u_n) - \frac{1}{2}u_n$ therefore yields

$$A[u] = \frac{3}{2}D[u] + \frac{1}{4}U[u], \quad E[u] = \frac{3}{2}D[u] + b_0U[u]. \quad (3.4)$$

The matrix of D has diagonal 2 and nearest off-diagonal entries -1 . For $\theta_j = j\pi/L$, the addition formula gives

$$2v_j(n) - v_j(n-1) - v_j(n+1) = (2 - 2\cos\theta_j)v_j(n).$$

The boundary values vanish, and $v_j(1) > 0$ for $1 \leq j < L$, so these vectors are nonzero. Their eigenvalues increase strictly because $0 < j\pi/(2L) < \pi/2$. Symmetry of the matrix makes eigenvectors at distinct eigenvalues orthogonal. There are $L-1$ of them in a space of dimension $L-1$, proving completeness. Expanding any u in this basis makes its Rayleigh quotient a weighted average of the eigenvalues, proving (3.3). $\square$

Thus the form on one fixed box is nonnegative precisely when

$$\alpha \leq \frac{1}{4} + \beta q_0 + 6 \sin^2 \left(\frac{\pi}{2L} \right). \quad (3.5)$$

Strict inequality gives positive definiteness; equality gives the one-dimensional kernel spanned by v_1 . If the inequality is reversed, amplitude scaling makes the unrestricted quadratic energy unbounded below, although its unit-norm Rayleigh quotient still has a finite minimum. The boxes $L = 0, 1$ have no nonzero interior modes and are excluded from the eigenvalue assertion.

Theorem 3.2 (Sharp bound uniform in box size). *For constant confinement,*

$$\inf_{L \geq 2} \inf_{u \neq 0} \frac{E[u]}{U[u]} = b_0. \quad (3.6)$$

No nonzero finite-box vector attains this infimum. A lower bound $E \geq \gamma U$ holds on every finite box if and only if $\gamma \leq b_0$.

Proof. Equation (3.4) gives $E \geq b_0 U$. Equality on a nonzero vector would require $D = 0$, hence a constant vector with zero endpoints, a contradiction. For sharpness take $u_n = 1$ at all interior sites. Then $U = L-1$, $D = 2$, and

$$\frac{E[u]}{U[u]} = b_0 + \frac{3}{L-1}. \quad (3.7)$$

For every $\epsilon > 0$, an integer $L > 1 + 3/\epsilon$ gives a quotient below $b_0 + \epsilon$. No larger uniform bound can hold. $\square$

There are three distinct regimes. If $b_0 > 0$, a positive lower bound is uniform in all boxes. If $b_0 = 0$, every fixed finite box is positive but there is no positive bound uniform in box size. If $b_0 < 0$, sufficiently large boxes have negative modes: the explicit slab is negative whenever $L > 1 + 3/(-b_0)$. Slabs establish sharpness; they are generally not the lowest sine eigenvectors on a fixed box.

In the square-parameter convention without confinement, all-box nonnegativity is $|\kappa| \leq 1/2$. Restricting $\kappa \geq 0$ changes this to $0 \leq \kappa \leq 1/2$, not to the single selected value $1/2$. With constant confinement it becomes $\kappa^2 \leq 1/4 + \mu^2 q_0$. For example, taking $\kappa^2 = \mu^2 = q_0 = 1$ gives $E = (3/2)D + U/4 \geq U/4$ despite $\kappa = 1$. Positive confinement therefore rules out a confinement-independent wall at $\kappa = 1/2$.

For variable $q_n \geq q_0$ and $\beta \geq 0$, the same identity gives $E \geq b_0 U$. This is optimal uniformly over the entire class $q \geq q_0$, because the class contains the constant profile. It need not be sharp for a specified nonconstant profile, and a sign assumption on β is essential for this comparison.

4 Why the scalar source action does not identify the discrete form

The exact discrete theorem does not establish that its energy is a physical fluctuation energy. We can test the proposed identification directly against the repository kinetic action, rather than assume that the repeated occurrence of $1/4$ supplies it. The source paths and their pinned hashes are recorded with the I14 computations [4].

On a fixed flat spatial background consider the radial functional

$$\mathcal{F}[\phi] = 4\pi \int r^2 \Lambda^4 f(K) dr, \quad K = \frac{\phi_r^2}{2\Lambda^4}, \quad (4.1)$$

with $\Lambda > 0$ a scale. The G155 source specifies, for $y = \sqrt{K} > 0$,

$$f(K) = y^2 - 2\log(1+y) - \frac{2}{1+y} + 1, \quad f_K = \mu(y) = \frac{y(y+2)}{(1+y)^2}. \quad (4.2)$$

The symbol $\mu(y)$ here is the source's interpolation function; it is unrelated to the confinement parameter in Section 3.

For $\phi = \phi_0 + \epsilon h$, direct Taylor expansion shows that the coefficient of ϵ^2 is

$$\mathcal{Q}[h] = 2\pi \int r^2 a_r(r) h_r^2 dr, \quad a_r = f_K + 2K f_{KK}. \quad (4.3)$$

Indeed, the first and second perturbations of K are $\phi_{0,r} h_r / \Lambda^4$ and $h_r^2 / (2\Lambda^4)$, respectively. Substitution in f gives (4.3); the full second variation is twice $\mathcal{Q}$. In this fixed-background calculation, shift symmetry produces a derivative-only quadratic form in h .

For (4.2),

$$a_r(y) = \mu(y) + y\mu'(y) = \frac{y(y^2 + 3y + 4)}{(1+y)^3} > 0. \quad (4.4)$$

Take the proposed logarithmic profile $\phi_0 = C \log r$, with $C > 0$, and write $\ell = C/(\sqrt{2}\Lambda^2)$, so $y = \ell/r$. With $r = e^t$,

$$\frac{\mathcal{Q}[h]}{2\pi} = \int w(t) h_t^2 dt, \quad w(t) = r a_r(\ell/r) = \ell \frac{y^2 + 3y + 4}{(1+y)^3}. \quad (4.5)$$

In the deep regime $y \rightarrow 0$, this weight tends to the constant 4ℓ . It does not tend to the canonical radial weight proportional to e^t .

For any positive weight, the unit-weight variable $v = \sqrt{w} h$ gives

$$\frac{\mathcal{Q}[h]}{2\pi} = \int (v_t - sv)^2 dt = \int (v_t^2 + (s^2 + s')v^2) dt, \quad s = \frac{1}{2}(\log w)', \quad (4.6)$$

under compact support or vanishing endpoint terms. A constant radial a_r has $w \propto e^t$, $s = 1/2$, and potential $1/4$; by contrast, $a_r \propto r^{-1}$ has constant w and potential zero. More generally, $a_r \propto r^{-p}$ gives potential $(1-p)^2/4$. The weight, rather than the mere appearance of a logarithmic profile, determines this number.

The exact interpolant gives

$$\begin{aligned} s(y) &= \frac{y(y^2 + 4y + 9)}{2(1+y)(y^2 + 3y + 4)}, \\ V_{\text{dress}}(y) &= s(y)^2 - y \frac{ds}{dy} \\ &= \frac{y(y^5 + 8y^4 + 42y^3 + 64y^2 + 17y - 72)}{4(1+y)^2(y^2 + 3y + 4)^2}. \end{aligned} \quad (4.7)$$

It tends to zero as $y \rightarrow 0$ and to $1/4$ as $y \rightarrow \infty$. It is not a constant $1/4 - \kappa^2$. A locally negative value of this dressed potential also does not imply a negative mode, since the original form is an integral of positive weighted squares.

There is a second obstruction: the proposed logarithmic profile is not an exact source-free background for the full interpolant. Its radial flux is

$$r^2 \mu(y) \phi_{0,r} = C\ell \frac{y+2}{(1+y)^2}, \quad r \frac{d}{dr} (r^2 \mu(y) \phi_{0,r}) = C\ell \frac{y(y+3)}{(1+y)^3} > 0. \quad (4.8)$$

Source-free equilibrium requires constant flux. The leading approximation $\mu(y) = 2y$ does give constant flux $2C\ell$, so the profile is compatible with that approximate equation, but the exact and approximate backgrounds must be distinguished.

Finally, another cited repository source uses $f(K) = K - 1/(1+K)$ with a different invariant convention. Its derivative at zero is 2, whereas the derivative in (4.2) tends to zero. A sign convention alone cannot identify these two functions. The related isothermal-fluid calculations also involve a different displacement, norm, and endpoint problem; local solutions of an ordinary differential equation do not determine the spectrum of a fixed self-adjoint domain.

These calculations concern a spatial functional on a fixed background. They do not decide the time kinetic sign or the stability of a fully coupled matter-gravity theory. What they do establish is the failure of the proposed direct map to (3.1). A physical application requires a specified action, its actual background, the constrained quadratic dynamics, its Hilbert norm and endpoints, and a mesh and parameter map. Selection of a unique coupling further requires a physical saturation condition beyond nonnegativity.

5 Spectral bounds for the $SU(N)$ lattice Hamiltonian

5.1 Hamiltonian, normalization, and Casimir minimum

Fix $d \geq 2$, $N \geq 2$, $x = g^2 > 0$, and $0 \leq b_N \leq 2N$, at lattice spacing one. Each positive lattice link has Hilbert space $L^2(SU(N))$ with normalized Haar measure. Reverse links are inverses, not independent factors. Let C_l be the positive Casimir with generator normalization $\text{Tr}(t_a t_b) = \delta_{ab}/2$, and put

$$H_N(x) = \frac{x}{2} \sum_l C_l + \frac{b_N}{x} \sum_p (1 - T_p), \quad T_p = \frac{\text{Re Tr } U_p}{N}. \quad (5.1)$$

The trace is a real multiplication operator with $-1 \leq T_p \leq 1$. The physical space consists of vectors invariant under all vertex gauge transformations, with no external charges. The repository

normalization $b_N = 2$, as well as $b_N = N$ and $b_N = 2N$, is covered by the assumptions. For finitely many links the electric Casimir sum has compact resolvent; bounded magnetic multiplication preserves self-adjointness on its domain and compact resolvent.

Lemma 5.1 (Minimum nonzero Casimir). *The least nonzero Casimir eigenvalue in $L^2(\text{SU}(N))$ is*

$$C_F = \frac{N^2 - 1}{2N} \geq \frac{3}{4}. \quad (5.2)$$

Proof. With roots of squared length two, the fundamental weights have Gram matrix

$$(\omega_i, \omega_j) = \frac{\min(i, j)(N - \max(i, j))}{N} > 0.$$

This follows by writing $\omega_i = \sum_{k \leq i} e_k - (i/N) \sum_{k \leq N} e_k$ in the hyperplane of zero coordinate sum. An irreducible representation has highest weight $\lambda = \sum_i a_i \omega_i$, with a_i nonnegative integers, and $C_2(\lambda) = (\lambda, \lambda + 2\rho)/2$, where $\rho = \sum_i \omega_i$. If $\lambda \neq 0$, choose $a_i \geq 1$ and write $\lambda = \omega_i + \nu$ with ν dominant. Then

$$C_2(\lambda) - C_2(\omega_i) = \frac{1}{2}(\nu, \nu) + (\nu, \omega_i + \rho) \geq 0.$$

Moreover,

$$C_2(\omega_i) = \frac{i(N-i)(N+1)}{2N} \geq C_F,$$

because $i(N-i) - (N-1) = (i-1)(N-i-1) \geq 0$. The fundamental representation attains equality. Peter–Weyl decomposition supplies all these sectors, so no representation cutoff is involved. $\square$

5.2 An untruncated single-plaquette bound

On an open four-edge square, gauge fixing a spanning tree leaves one holonomy modulo conjugation. The physical Hilbert space is therefore the space of class functions on $\text{SU}(N)$, with irreducible characters as an orthonormal basis. This standard reduction is also described in [3]. Each of the four link Casimirs acts on a character χ_R by $C_2(R)$, so

$$H_E \chi_R = 2x C_2(R) \chi_R, \quad E_0(H_E) = 0, \quad E_1(H_E) = 2x C_F. \quad (5.3)$$

Eigenvalues here and below are ordered with multiplicity.

Theorem 5.2 (Single plaquette). *For the open square with Hamiltonian (5.1),*

$$E_1(H) - E_0(H) \geq 2x C_F - \frac{b_N}{x} \geq x \frac{N^2 - 1}{N} - \frac{2N}{x}. \quad (5.4)$$

In particular, for every $N \geq 2$ and $x \geq 2$ this gap is at least one.

Proof. The magnetic operator is nonnegative. The full min–max principle and (5.3) give $E_1(H) \geq 2x C_F$. The normalized constant function is an admissible trial vector. Haar integration of the nontrivial fundamental character vanishes, so its expectation is $\langle 1, H1 \rangle = b_N/x$, giving $E_0(H) \leq b_N/x$. Subtract the two inequalities. Finally, for $x \geq 2$,

$$x \frac{N^2 - 1}{N} - \frac{2N}{x} \geq \frac{N^2 - 2}{N} = N - \frac{2}{N} \geq 1.$$

$\square$

The proof acts on the full physical character space. The constant electric vacuum is used only as a trial vector: when $b_N > 0$, $H1 = (b_N/x)(1 - T)$ is nonconstant. In particular, subtracting its expectation does not subtract the interacting ground energy. At $x = 2$, the bound using $b_N = 2$ is $2N - 2/N - 1$, giving 2 for $\text{SU}(2)$ and $13/3$ for $\text{SU}(3)$. These values, and the weaker values 1 and $7/3$ obtained from $b_N \leq 2N$, are lower bounds, not exact gaps.

5.3 A common strong-coupling window for all N and open volumes

We use the following external input from Yarotsky [2], Theorems 1–3, pp. 2–4. At each site of $\mathbb{Z}^d$ allow an arbitrary, possibly infinite-dimensional Hilbert space with a nonnegative self-adjoint onsite operator, simple zero vacuum, and gap at least one. Add bounded self-adjoint interactions ϕ_z supported in translates of one fixed finite set S . There are positive $c_1(S), c_2(S)$ such that, if $\epsilon_* := \sup_z \|\phi_z\| < c_1$, finite empty-boundary restrictions have a unique ground vector and gap at least $1 - c_2\epsilon_*$. The canonical thermodynamic state and its GNS Hamiltonian exist, with the same gap estimate. The constants depend only on S , not on site dimension or the upper onsite spectrum. The cited paper reviews the proof in Section 2; its cluster expansion is an external theorem, not reproduced or Lean-formalized here.

Theorem 5.3 (Uniform open-lattice strong-coupling bound). *For each fixed $d \geq 2$ there exists a finite X_d , independent of N , b_N , and finite open volume, such that for $N \geq 2$, $0 \leq b_N \leq 2N$, and $x \geq X_d$, the following statements hold.*

1. *Every finite open subgraph of the cubic lattice, with selected elementary plaquettes, has a unique physical ground vector and*

$$H - E_0 \geq \frac{3x}{16} \quad \text{on its physical vacuum complement.} \quad (5.5)$$

If that complement is zero, the spectral-exclusion statement is vacuous.

2. *The canonical GNS Hamiltonian for the homogeneous infinite interaction has a simple zero eigenvalue and*

$$\text{Spec}(H_\infty) \subset \{0\} \cup [3x/16, \infty). \quad (5.6)$$

The same conclusions hold on the cyclic space generated by local gauge-invariant observables.

Proof. Group the d positive outgoing links at z into the site space

$$\mathcal{H}_z = L^2(\text{SU}(N)^d), \quad h_z = C_F^{-1} \sum_{i=1}^d C_{(z,i)}.$$

The product of constant functions is its unique vacuum, and (5.2) makes the onsite gap exactly one. Divide energies by $s = xC_F/2$ and remove the scalar magnetic constant. With $m_d = \binom{d}{2}$, group plaquettes at their lower anchor:

$$\phi_z = -\frac{2b_N}{x^2 C_F} \sum_{i < j} T_{(z,ij)}. \quad (5.7)$$

The four link factors lie in the site blocks $z, z + e_i, z + e_j$, so all these interactions have the same allowed support $S = \{0, e_1, \dots, e_d\}$. Since $\|T_p\| \leq 1$,

$$\epsilon_* \leq \frac{2b_N m_d}{x^2 C_F} \leq \frac{A_d}{x^2}, \quad A_d = \frac{32}{3} \binom{d}{2}. \quad (5.8)$$

Indeed, $b_N/C_F \leq 4N^2/(N^2 - 1) \leq 16/3$. Thus the small parameter is uniform in N , despite the growing representation content of each site.

Choose

$$X_d = \max \left\{ 1, \sqrt{\frac{2A_d}{c_1(S)}}, \sqrt{2c_2(S)A_d} \right\}. \quad (5.9)$$

For $x \geq X_d$, we have $\epsilon_* \leq c_1/2 < c_1$ and $c_2\epsilon_* \leq 1/2$. The external theorem gives a normalized gap at least $1/2$. Restoring units yields $xC_F/4 \geq 3x/16$ and subtracts the actual interacting ground energy. We next verify the boundary and gauge statements needed to apply this estimate in the announced spaces.

For a finite open subgraph, retain $L^2(\text{SU}(N))$ only on its actual positive links and replace each absent factor by $\mathbb{C}$. An empty site has $h_z = 0$ on a one-dimensional space; the gap condition on its zero-dimensional vacuum complement holds vacuously. Let (5.7) include only selected elementary plaquettes. Pad the finite site set so that it contains $z + S$ for every active anchor. The resulting inhomogeneous model is exactly an allowed finite restriction with the same interaction range and norm bound. Dummy factors introduce neither link degrees of freedom nor boundary charges.

Every term of (5.1) commutes with the finite product of vertex gauge groups, including all endpoints of present links. The unique unconstrained ground vector spans a continuous one-dimensional representation of that product. A finite product of $\text{SU}(N)$ groups has no nontrivial continuous characters, so this vector is gauge invariant. The physical space reduces the Hamiltonian and contains its ground vector. Restricting the excited-space quadratic-form inequality proves (5.5).

For the infinite system, use the source's fixed homogeneous empty-boundary exhaustion: retain a grouped interaction when all of its declared support lies inside the site volume. This need not coincide with retaining every individual boundary plaquette of a conventional finite box, but it exhausts the homogeneous interaction. The external theorem gives its canonical state ω_∞ and GNS triple $(\mathcal{H}_\infty, \pi, \Omega)$, with (5.6). Changing padded finite models is not used to assert this thermodynamic existence.

It remains to justify the physical cyclic restriction. For a finite vertex set F , let $\mathcal{E}_F$ be Haar averaging on local operator algebras over the gauge transformations at F . It is a normal conditional expectation in the underlying finite tensor product. Gauge-invariant finite-volume vacua satisfy

$$\omega_\Lambda(B^* \mathcal{E}_F(A)) = \omega_\Lambda(\mathcal{E}_F(B)^* A), \quad \|\mathcal{E}_F(A)\Omega_\Lambda\| \leq \|A\Omega_\Lambda\|.$$

For fixed local A, B, F these identities pass to the thermodynamic state. Consequently

$$P_F \pi(A) \Omega = \pi(\mathcal{E}_F(A)) \Omega$$

defines an orthogonal projection on the GNS space. For increasing finite sets F exhausting all vertices the projections decrease and converge strongly to a projection P . Averaging any local operator over all vertices incident to its links already makes it globally gauge invariant. It follows, by applying the projections to the dense set of local GNS vectors, that

$$\text{Ran } P = \overline{\{\pi(A)\Omega : A \text{ local and gauge invariant}\}}. \quad (5.10)$$

Finite-volume resolvents commute with gauge averaging. The weak resolvent matrix-element convergence in Yarotsky's Theorem 3 passes this identity to P_F and the limiting resolvent, and then to P . Thus (5.10) reduces H_∞ , contains Ω , and inherits its simple vacuum and its gap. This argument does not require assuming norm continuity of the gauge action on every local bounded operator. $\square$

The constant (5.9) is an existence threshold. No numerical values for c_1, c_2 have been extracted, so the theorem does not establish $X_d \leq 2$, or any other proposed numerical onset. It is also a fixed-spacing, fixed- d assertion. A limit with fixed 't Hooft coupling Nx and growing N eventually leaves

the assumed window. Periodic tori are not included by the open-subgraph argument. The simple vacuum in the canonical GNS representation does not assert uniqueness among all infinite-volume ground-state representations. No continuum scaling, confinement observable, or relation to the scalar theory in Section 4 is supplied by this theorem.

6 Benchmarks, reproducibility, and certification limits

6.1 Numerical checks

The radial implementation generates equilibria rather than importing stellar evolution profiles. For the $n = 3$ models it uses $G = \rho_c = \text{Lane–Emden length scale} = 1$, $P_c = \pi$, and $c^2 = \pi/s$ when $s = P_c/(\rho_c c^2) > 0$. It integrates the TOV or Lane–Emden equations adaptively and assembles continuous piecewise-linear Rayleigh–Ritz forms on a uniform radial mesh with eight-point Gaussian quadrature per cell. Endpoints are not quadrature nodes, and no artificial zero-displacement condition is imposed at the surface. Calculations use binary64 arithmetic, not interval enclosures.

The generated Newtonian $n = 3$ solution has radius approximately 6.89684861937 and Lane–Emden mass moment 2.01823595097. At $\Gamma = 4/3$ it reproduces the homologous zero mode. At $\Gamma = 1.5$, the 256-cell value in Table 2 is below the homologous quotient; the latter exceeds it by approximately 20.43%. Nested 32-, 64-, 128-, and 256-cell trial spaces give decreasing computed Ritz values. In exact arithmetic, nested Rayleigh–Ritz spaces provide upper bounds and monotonicity; the floating calculation checks their numerical realization and does not produce a certified lower bound.

Benchmark	Comparison value	Computed value
$n = 3$, Newtonian, $\Gamma = 1.5$	$Q_N[1] = 0.5129469320$	Ritz 0.4259235109
$n = 3$, TOV, $s = 0.001$, selected Γ	Homology $+2.41886 \times 10^{-7}$	Ritz -2.42086×10^{-7}
Single SU(2) plaquette, $x = 2$, $b_N = 2$	Lower bound 2	Ritz gap 3.1138638115
Single SU(3) plaquette, $x = 2$, $b_N = 2$	Lower bound 13/3	Ritz gap 5.1762835133

Table 2: Selected checks, in the code’s stated units. The TOV row uses $\Gamma = 1.335967888598$ and compactness 0.002325278673. Radial Ritz values use 256 cells. Character-matrix gaps are differences of finite Ritz levels and are not certified lower bounds on the full gap. The analytic bounds are proved independently of these computations.

The TOV row is a numerical illustration of a positive homologous value coexisting with negative work on another admissible trial space. Its small sign difference has not been interval-certified. The analytic comparison direction and the residual obstruction (2.7) do not rely on that numerical sign. Independent assembly of (2.14) and (2.17) agrees in the tested model to about 2.36×10^{-12} in the largest matrix-entry difference.

For I14 the numerical check constructs the original rectangular first-difference operator $Bu = (u_{n+1} - 3u_n/2)_n$, then forms $B^T B$ and the scalar shifts. Thirty matrices, using $L \in \{2, 3, 4, 8, 31, 128\}$ and five parameter triples, agree with (3.2) to at most 6.22×10^{-15} . Exact symbolic checks separately verify the source-action derivatives and flux obstruction. These tests check constants and indexing; completeness and sharpness follow from the proofs in Section 3.

The I15 script performs exact rational Casimir checks over a declared finite family and constructs SU(2) and SU(3) character truncations at multiple cutoffs. Its 159 recorded assertions cover normalization, the displayed inequalities, cutoff comparisons, and failure of bare-vacuum subtrac-

tion. Finite diagonalizations are not used to infer a universal Casimir minimum or an infinite-space lower gap bound.

6.2 Formal coverage

The three Lean sources compile in the recorded project environment. Their printed theorem dependencies contain only `propext`, `Classical.choice`, and `Quot.sound`; no custom axioms or proof placeholders were introduced in these certificates. This describes the checked declarations, not a formal verification of the complete manuscript.

For I13, the existing scalar proof bodies were retained and their interpretation corrected. The radial differential operator, domain analysis, and post-Newtonian expansion remain analytic arguments in prose. For I14, seventeen added declarations cover sharp uniform inequalities, confinement, finite positivity, sine recurrence and energy identities, and nonzero norms. The completeness argument and exact finite spectral minimum are supplied in Section 3, not claimed as fully formalized declarations. For I15, six added declarations cover scalar comparisons, the Casimir floor expression, local perturbation normalization, and energy rescaling. Peter–Weyl theory, min–max, the external stability theorem, and the gauge/GNS transfer are not claimed as Lean-formalized.

6.3 Archive layout and reproduction

Repository-relative source paths are used throughout the archive. The consolidated directory is

```
real_research/reviews/spectral_spine_closure_2026_09_22/.
```

Its `REPORT.md` records the combined scope; the principal derivations are `i13/REPORT.md`, `i14/DERIVATION.md`, `i14/SOURCE_BRIDGE.md`, and `i15/PROOF.md`. The audit baseline is commit `e3af62a453ca6625db8428f6d32957b27a7f1331`. This baseline identifies the starting inputs, not a claim that it already contains the later results. Final file hashes and executed verification records identify the checked artifacts. The publication archive should be used with its matching source files rather than an unpinned future branch.

The numerical and symbolic entry points are `i13/radial_bridge.py`, `i13/symbolic_identities.py`, `i14/verify.py`, and `i15/verify.py`, relative to that directory. Their adjacent contracts, results, manifests, and preserved standard output specify commands, dependencies, parameter ranges, resource limits, input hashes, and exit status. For example, from the repository root the radial checks can be repeated as

```
python3 real_research/reviews/spectral_spine_closure_2026_09_22/
i13/radial_bridge.py -output /tmp/i13_results.json
```

where the displayed line break is typographical and the script path is a single shell argument. The symbolic script accepts its output JSON path as a positional argument. For all checks, the stored manifest is the authoritative command record; paths can be redirected to a local output directory without changing the mathematical inputs.

The Lean files are in `fable_independent_2026/lean_2026/`:

```
I13_bhstar_pulsation.lean
I14_phantom_vacuum_wall.lean
I15_suN_lattice_gap.lean
```

From that directory each is checked with `lake env lean` followed by its filename. The archive records successful runs, with 17 numerical assertions and four symbolic identities for I13, the

I14 matrix and symbolic checks, and the I15 assertion suite. The verification directory preserves combined compiler and manifest summaries. Successful execution and matching hashes establish provenance and the recorded checks; they do not independently prove every analytic step.

The source records pin Saio et al. to version 2406.18040v1 and Yarotsky to `math-ph/0411042v1`. They specify theorem and equation locations and the application boundaries. Those records do not claim that complete source PDFs were retained or hashed locally. The mathematical claims use the pinned source versions listed in the bibliography. Public repository links are provided in References [4, 5, 6].

7 Consequences and remaining mathematical tasks

The radial calculation supplies the missing variational comparison and fixes the inertia and relativistic coefficient conventions. A stellar mass cutoff still requires an actual equilibrium sequence, appropriate equation of state, relevant inner and outer boundaries, and either the exact relativistic test or a bounded post-Newtonian remainder. The generated polytropes are useful benchmarks, not MESA or GENEC profile data and not a black-hole-envelope mass sequence.

The discrete calculation closes the spectrum of the specified constant-confinement model, including the distinction between a finite box and a bound uniform in box size. The action calculation then identifies why that theorem cannot presently select a physical acceleration coupling: the cited source gives a different radial weight, no independently derived negative κ^2 potential, and a logarithmic profile that is only an approximate background for its full interpolant. Replacing an algebraic coefficient does not resolve those operator-identification requirements.

The lattice results provide two actual gap comparisons, one elementary and one using an established external theorem. Both refer to the shifted interacting ground state. The open-volume theorem is uniform in gauge rank because the local Casimir normalization controls the interaction norm independently of N . Evaluating a useful explicit X_d , reaching $x = 2$ uniformly in arbitrary open volume, and analyzing any continuum limit are separate tasks. None is established by the positivity of the single-plaquette scalar expression.

Together the examples show the practical role of a certificate’s interface: one must say which quadratic form is represented, in which norm and domain, and whether the certified expression is an upper trial value, a lower bound, or an exact eigenvalue. Those distinctions determine which physical conclusions the mathematics can support.

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AI assistance and review status

OpenAI Codex assisted with drafting, symbolic and numerical code, source inspection, and mathematical consistency checks for this manuscript and its accompanying artifacts. Separate AI-assisted checks were used for some operator and source mappings. Such checks are not independent human

peer review, and the disclosure does not imply that every computation or derivation has been manually reviewed by the author. The executable records and stated limits identify the evidence being presented. No AI system is listed as an author.

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- [5] C. P. Zimmerman, *I13, I14, and I15 Lean certificates*, [fable_independent_2026/lean_2026/](#), accompanying repository sources. [Public repository: Lean source directory](#). Compilation records and final source hashes appear in Reference [4].
- [6] C. P. Zimmerman, *Project source actions and prior spectral calculations*, [deepseek_push/G155_sourced_eq.py](#), [deepseek_push/G081_equilibrium_stability.py](#), [deepseek_push/WAVEBOARD.md](#), and [deepseek_push/THE_THEORY.md](#). [Public repository: inspected project sources](#). These are provenance sources for the proposed physical identification; the limitations established in Section 4 apply.